

PP Technical Data Sheet (TDS)

Polypropylene (PP) is a durable, high toughness and exceptional fatigue resistance. It has good resistance in chemical, thermal and electrical. It is one of the commonly used plastic on earth such as electrical components and prototyping.

It is commonly applied in field such as functional prototypes, guides, sleeves, and protective cases.

Mechanical Properties	Injection Molding	
	Typical Value	Test Method
Tensile Modulus	380MPa	ISO 527
Tensile Stress at Yield	13MPa	ISO 527
Tensile Stress at Break	No break within testing range	ISO 527
Elongation at Yield	13.5%	ISO 527
Elongation at Break	>210%	ISO 527
Flexural Strength	13MPa	ISO 178
Flexural Modulus	360MPa	ISO 178
Izod Impact Strength (at 23 °C)	-	-
Charpy Impact Strength (at 23 °C)	10.5 kJ/m ²	ISO 179
Hardness	54 (Shore D)	ISO 868

Electrical Properties	Value	Test Method
Volume Resistivity	> 10 ¹⁶ Ohm.cm	(Typical Value)
Dissipation factor (at 1 MHz)	-	-
Dielectric constant (at 1 MHz)	-	-

Thermal Properties	Typical Value	Test Method
Melt Mass-Flow Rate (MFR)	20 g/ 10 min	ISO 1133 (23°C, 2.16 Kg)
Heat Detection (at 0.455 MPa)	-	-
Heat Deflection (at 1.82MPa)	-	-

Vicat Softening Temperature at 5N	114°C	ISO 306
Glass Transition	-	-
Coefficient of Thermal Expansion	-	-
Melting Temperature	132°C	DSC
Thermal Shrinkage	-	-

Other Properties	Value	Test Method
Specific Gravity	0.89	ISO 1183
Flame Classification	-	-
Haze (1 mm)	28%	ISO 14782
Glass	90%	ASTM D523

Print Recommendation	
Nozzle Temperature	230 -260 °C
Bed Temperature	90 -110 °C
Print Speed	30-70 mm/s
Chamber Temperature	60-80 °C
Cooling Fan	0-50%